

Subject Description Form

Subject Code	AAE5105
Subject Title	Fleet Management and Aviation Sustainability
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. advanced knowledge in airline fleet sizing, fleet assignment, flight scheduling, aircraft assignment, and aircraft routing; 2. the advanced knowledge of the major operational, technical and inventory support functions of the airline industry; and 3. the advanced emission mitigation strategy and sustainable aviation operations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. design and develop mathematical modelling in resolving airline fleet, crew pairing and aircraft routing problems; b. design and develop proper airline resource planning in a profitable manner; c. apply various strategies and techniques to optimise and implement aircraft maintenance programmes; d. understand and apply the various inventory support models to the airline; e. evaluate the impact of aviation emission and its mitigation strategy; and f. determine airline solutions contributing to the societal, economic and global environment factors.
Subject Synopsis/ Indicative Syllabus	Operations management, fleet and crew management and flight route management: Airline fleet management, crew management, aircraft routing and sustainability; Airline network design and route planning; Aircraft life cycle and associated legislation; Human resource management: crew pairing and rostering management. Operational and technical Support: Technical support functions in maintenance, repair and overhaul; development, implementation and optimisation of maintenance programmes; inventory support models and their implementation; Sustainable aviation: Carbon budgets for aviation; Environmental technology and the future of flight; Aircraft fuel consumption calculation and reduction; Environmental impact of aviation emission; Sustainable

	aviation system.							
Teaching/Learning Methodology	Teaching is conducted through lectures and assignments. The basic knowledge, research methodology and theoretical models will be introduced.							
	The understanding of how to address and formulate problems by using mathematical programming, data analytics, and operations research techniques is emphasised. Research methodologies, such as data analytics and mathematical modelling skills, are taught in class as well as the related real-life scenarios using data to enhance their research abilities.							
	Teaching/Learning Methodology	Outcomes						
		a	b	c	d	e	f	
Lecture	√	√	√	√	√	√		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Assignment	30%	√	√	√	√	√	
	2. Course Project	20%			√	√	√	√
	3. Final examination	50%	√	√	√	√	√	√
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × Continuous Assessment + 0.5 × Final Examination The continuous assessment (50%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via assignments and a course project. The final examination (50%) will also be considered to assess the students’ learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lecture					39 Hrs.		
	Other student study effort:							
	▪ Self-study / preparation					66 Hrs.		

	Total student study effort	105 Hrs.
Reading List and References	1. Abdelghany, A., & Abdelghany, K. (2016). Modeling applications in the airline industry. Routledge. 2. Bazargan, M. (2016). Airline operations and scheduling. Routledge. 3. Bridger, R. (2013). Plane truth: Aviation's real impact on people and the environment. 4. Budd, L., Griggs, S., & Howarth, D. (2013). Sustainable aviation futures. Emerald Group Publishing. 5. Clark, P. (2017). Buying the big jets: fleet planning for airlines. Taylor & Francis. 6. Walker, T., & Bergantino, A. S. (2020). Sustainable Aviation. Palgrave Macmillan. 7. Wu, C.-L. (2016). Airline operations and delay management: insights from airline economics, networks and strategic schedule planning: Routledge. 8. Kinnison, Harry A., and Tariq "Terry" Siddiqui (2013). "Aviation Maintenance Management. 2nd ed. New York: McGraw-Hill Education.	

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